EE3032 Introduction to VLSI – Midterm

Fall 2011

1. (20 points) Complete the following problems.

(a) The layer density rule should be met when you design a nano-scale IC. Why? (2 points)

(b) What is the antenna effect? (2 points)

(c) Draw a CMOS transistor-level schematic realizing $f = \overline{xy + (u+v)w}$. (4 points)

(d) List four properties of static CMOS logic. (4 points)

(e) Draw a CMOS inverter and label the drain and source of its PMOS and NMOS. (2 points)

(f) List the three major components of load capacitance of a CMOS logic gate in a CMOS circuit. (3 points)

(g) How do you size the N and P MOS of an inverter such that its output voltage is $V_{DD}/2$ when its input is $V_{DD}/2$? (3 points)

2. (20 points) An inverter with output load C_L, operated at frequency f under supply V_{DD}, has switching power $fC_LV_{DD}^2$. If input $t_r = t_f = \tau$ is applied to an inverter with $\beta_n = \beta_p = \beta$, its short-circuit power can be estimated by the printed formula $P_{sc} = k\beta(V_{DD} - V_t)^3\tau f$, where k is a constant. Also $t_r = C_L/(\beta_p V_{DD})$ and $t_f = C_L/(\beta_n V_{DD})$.

(a) Figure 1 depicts an inverter chain driving C_L. Assume $\mu_n = 2\mu_p$. A W/L PMOS or NMOS has gate capacitance C_g, and a W/L NMOS has $\beta_n = \beta$. Drain/wire parasitics are neglected. A unit delay is $\tau_0 = C_g/(\beta V_{DD})$ ns. Calculate the dynamic power (switching and short-circuit) of the chain when the shown waveform is applied to its input at switching frequency f. (10 points)

(b) The minimum-size inverter has W/L NMOS and W/L PMOS, with total input capacitance C_g; its delay driving C_g is t_d. For brevity, assume $e = 2$ and neglect drain/wire capacitance. Design a buffer chain driving $C_L = 64C_g$ with minimized delay. Draw the chain, label transistor sizes, and explain. (10 points)

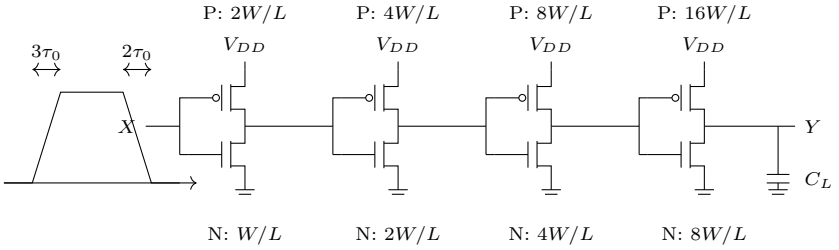
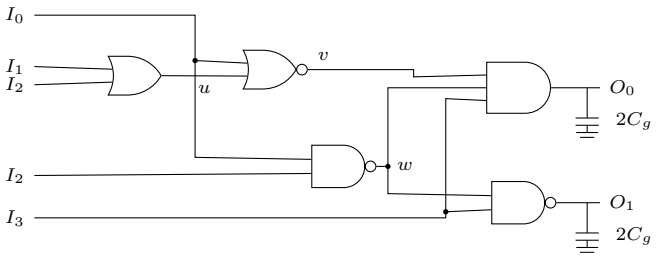


Figure 1: Inverter chain; PMOS/NMOS widths are 2/1, 4/2, 8/4, 16/8 times W.

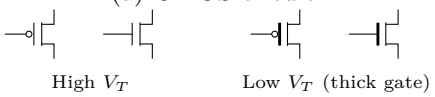
3. (20 points) Figure 2(a) depicts a CMOS logic circuit. An inverter has delay 1 unit, with $W_p/L_p = 2\,\mu\text{m}/0.18\,\mu\text{m}$, $W_n/L_n = 1\,\mu\text{m}/0.18\,\mu\text{m}$, and input capacitance C_g. All transistors in the other gates have the corresponding same sizes. Functional inputs are $\{I_0 I_1 I_2 I_3\} = \{0100, 0011, 1100, 1011\}$, periodically applied in sequence at frequency f under supply V_{DD}.

(a) A dual-V_T process can implement the circuit. Draw its transistor-level schematic using the symbols in Figure 2(b), such that leakage power is minimal without sacrificing speed. (10 points)

(b) Assume all logic gates in Figure 2(a) have the same delay. Does the circuit exhibit glitches? Explain. (10 points)



(a) CMOS circuit



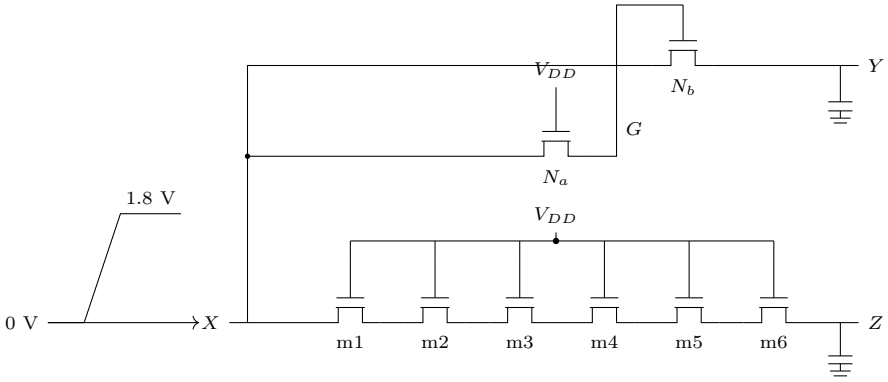
(b) High- and low-V_T symbols

Figure 2.

4. (20 points) Figure 3 shows a pass-transistor logic circuit. Assume $V_{DD} = 1.8$ V and NMOS threshold voltage 0.4 V.

(a) Draw Y and Z waveforms for the shown X input, label minimum and maximum voltages, and explain. (10 points)

(b) Neglect body effect and apply the same input. (i) Identify which of m1–m6 can operate in saturation and linear regions; explain. (5 points) (ii) Identify which can operate only in the linear region; explain. (5 points)



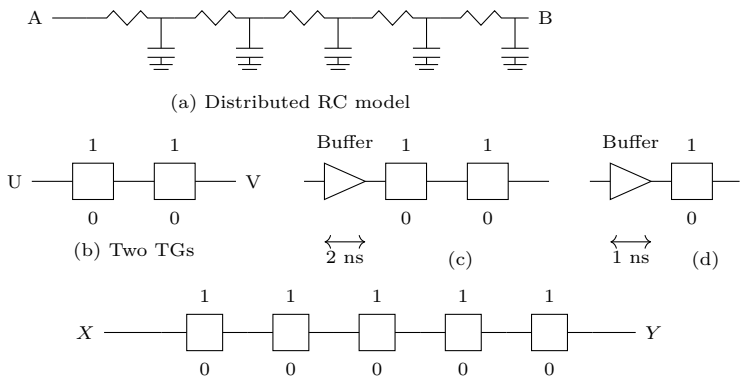
Source junction assumption approved by user:
the six-gate bus and N_a gate are tied to V_{DD}.

Figure 3: Pass-transistor circuit. N_a, N_b, G are added identification labels.

5. (20 points) Figure 4 depicts an interconnection RC model and different transmission-gate configurations.

(a) An interconnection is modeled as distributed RC; the delay of an $x\,\mu$m interconnection is estimated as kx^2, with constant k. What approach reduces interconnection delay? Explain. (5 points)

(b) The delay of two cascaded TGs in Figure 4(b) is 9 ns. A buffer driving two TGs / one TG has delay 2 ns / 1 ns, respectively, as shown in (c)/(d). (i) What is the delay of five TGs in (e)? (5 points) (ii) Insert the smallest number of buffers so delay is smaller than 30 ns; draw the circuit and calculate delay. (5 points) (iii) Insert buffers to minimize delay; draw the circuit and calculate delay. (5 points)



(e) Five cascaded TGs.

Figure 4: (a) Wire RC model, (b) two TGs, (c)/(d) buffered TGs, (e) five TGs.

6. (10 points) Figure 5 depicts a CMOS inverter process cross section. (a) Draw a circuit model of latchup. (5 points) (b) Explain latchup using the model in (a). (5 points)

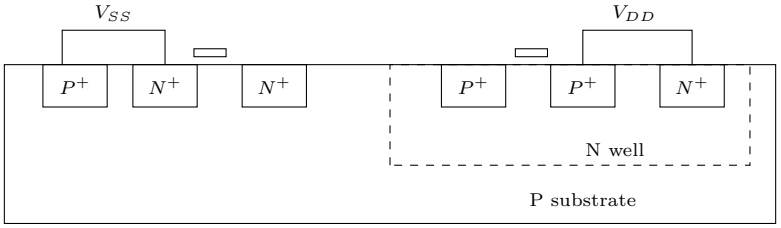


Figure 5: Process cross-section view of a CMOS inverter.

Archive note: Six source photograph panels cover the three-page dated Fall 2011 paper. Their unmodified JPEGs are preserved in original-exam.pdf. Printed question totals sum to 110 points; all original point labels are retained. The user approved the faint VDD-bus junction reconstruction in Figure 3 with this explicit assumption.